

Linear Equations: Guide

The big idea. A linear function $f(x) = mx + b$ changes by the **same amount** every time x goes up by one. That amount is the slope m , and it is the **rate of change**. Tables, graphs, equations, and word problems are four views of the same line.

IS IT LINEAR? FROM DATA TO A MODEL

- 1

Find each rate. For consecutive rows compute $\frac{\Delta y}{\Delta x}$. Do not assume the x -steps are equal.
- 2

Compare. Same rate every time means linear. One different rate means **not** linear.
- 3

Name the slope. That common rate is m , with units (dollars per hour, feet per second).
- 4

Find b . Plug a point into $y = mx + b$, or read $f(0)$ if the table has $x = 0$.
- 5

Write it. $f(x) = mx + b$ in function notation.
- 6

Check a second row before you trust the model.

Most common mistakes: (1) Declaring a table linear because y keeps going up. It must go up by a **constant rate**. (2) Sign errors on $x + 2$ vs $x - 2$ in point-slope. (3) In a word problem, putting the one-time fee with x and the per-unit rate as the constant. (4) Solving $f(x) = k$ by plugging k in for x . Plug in only when the input is given.

WHAT IS THE QUESTION ASKING?

THE QUESTION SAYS	IT MEANS	DO THIS
Find $f(3)$	Input is 3	Replace x with 3 and simplify.
Find x when $f(x) = 11$	Output is 11	Set $mx + b = 11$ and solve for x.
Where does it cross the axes?	Intercepts	x-int: set $y = 0$. y-int: set $x = 0$.
When are the two equal?	Intersection	Set the two expressions equal, solve x, then find y.
Parallel / perpendicular to...	Compare slopes	Parallel: $m_1 = m_2$. Perpendicular: $m_1 m_2 = -1$.

FORMULA BANK

SLOPE & RATE OF CHANGE Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$ Average rate $\frac{f(b) - f(a)}{b - a}$ For a line, the average rate of change is the same on every interval: it equals m.	FORMS OF A LINE Slope-int. $y = mx + b$ Point-slope $y - y_1 = m(x - x_1)$ Standard $Ax + By = C$ Standard form: $m = -\frac{A}{B}$.	FUNCTION NOTATION $f(3)$ is the output when the input is 3. $f(a) = k$ means the point (a, k) is on the graph. Zero of f: where $f(x) = 0$. It is the x-intercept.
PARALLEL & PERPENDICULAR Parallel $m_1 = m_2$ Perpendicular $m_2 = -\frac{1}{m_1}$ Parallel lines never meet. Perpendicular lines meet at a right angle.	SOLVING TWO LINES Set equal: if both are $y = \dots$, set the right sides equal. Substitute: put $y = \dots$ into the other equation. Always finish by finding both x and y.	INTERPRETING A MODEL $f(x) = mx + b$: m = change in output per one input unit. b = output when input is 0 (the start). Negative m means decreasing.

Eight problems that cover the exam

Same plan every time: rates, model, solve, check.

Walkthrough 1: Is the table linear?

CONSTANT RATE

 Table 1: $x = 0, 2, 5, 6$ and $y = 3, 7, 13, 15$.

 Table 2: $x = 0, 2, 5, 6$ and $y = 3, 7, 12, 15$.

1 RATES Table 1: $\frac{7-3}{2} = 2, \frac{13-7}{3} = 2, \frac{15-13}{1} = 2$.

2 COMPARE All equal 2: Table 1 is **linear**.

3 TABLE 2 $\frac{12-7}{3} = \frac{5}{3} \neq 2$. **Not** linear.

4 MODEL Table 1: $b = f(0) = 3$, so $f(x) = 2x + 3$

Walkthrough 2: Function notation

INPUT VS OUTPUT

$f(x) = -3x + 5$. Find $f(4)$, then find x when $f(x) = -7$.

1 F(4) Input is given: $-3(4) + 5 = -12 + 5 = -7$

2 EQUATION $-3x + 5 = -7$

3 SOLVE $-3x = -12$, so $x = 4$.

4 CHECK $f(4) = -7$ agrees with Step 1.

Walkthrough 3: A word problem

RATE AND START

A gym charges a \$30 sign-up fee plus \$18 per month. Write $T(m)$ and find when the total reaches \$246.

1 RATE 18 per month is the slope. 30 is the start.

2 MODEL $T(m) = 18m + 30$

3 SOLVE $18m + 30 = 246$, so $18m = 216$ and $m = 12$ months.

Walkthrough 8: Average rate of change

SAME FOR A LINE

$f(x) = 4x - 3$. Find the average rate of change from $x = 1$ to $x = 5$.

1 OUTPUTS $f(1) = 1$ and $f(5) = 17$.

2 RATE $\frac{17-1}{5-1} = \frac{16}{4} = 4$

3 NOTICE It equals the slope. For a line, every interval gives the same answer.

Walkthrough 4: Graph with intercepts

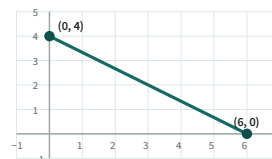
TWO AXIS POINTS

Graph $2x + 3y = 12$.

1 X-INT. $y = 0: 2x = 12$, so $(6, 0)$.

2 Y-INT. $x = 0: 3y = 12$, so $(0, 4)$.

3 DRAW Connect the two points. Slope $= -\frac{2}{3}$.



Standard form: use intercepts

Walkthrough 5: Perpendicular line

FLIP AND NEGATE

Write the line through $(-3, 4)$ perpendicular to $3x + 2y = 6$.

1 SLOPE $2y = -3x + 6$, so $y = -\frac{3}{2}x + 3$ and $m = -\frac{3}{2}$.

2 PERP. M Flip and negate: $m_{\perp} = \frac{2}{3}$.

3 SETUP $y - 4 = \frac{2}{3}(x + 3)$

4 SIMPLIFY $y - 4 = \frac{2}{3}x + 2$, so $y = \frac{2}{3}x + 6$

Walkthrough 6: Compare two plans

SET EQUAL

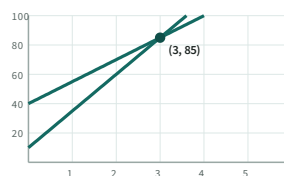
Plan A: $C = 40 + 15t$. Plan B: $C = 10 + 25t$. When do they cost the same? Which is cheaper afterward?

1 SET EQUAL $40 + 15t = 10 + 25t$

2 SOLVE $30 = 10t$, so $t = 3$.

3 COST $40 + 15(3) = 85$. They meet at $(3, 85)$.

4 AFTER B grows faster ($25 > 15$), so for $t > 3$ Plan A is cheaper.



Where the lines cross

Walkthrough 7: Substitution

ONE LINE IS SOLVED

Solve $y = 2x - 1$ and $3x + y = 9$.

1 PLUG IN $3x + (2x - 1) = 9$

2 SOLVE $5x = 10$, so $x = 2$.

3 FIND Y $y = 2(2) - 1 = 3$.

4 ANSWER The lines cross at $(2, 3)$. Check in $3x + y = 9$: $6 + 3 = 9$ ✓